1. Common Network Threats and Real-World Examples

Network threats are activities that can compromise the confidentiality, integrity, or availability of data and computer systems.

No.	Network Threat	Real-World Example	Mitigation
1	Phishing	A user receives a fake Instagram security message asking them to click a link and enter their login credentials.	Verify links and never enter passwords on suspicious websites.
2	DDoS (Distributed Denial-of-Service)	Attackers send a huge amount of traffic to an online service, making it slow or unavailable to legitimate users.	Use DDoS protection, traffic filtering, and rate limiting.
3	Man-in-the-Middle (MITM)	On an unsecured public Wi-Fi network, an attacker attempts to intercept communication between a user and an online service.	Use HTTPS, trusted networks, and a VPN when appropriate.
4	Malware	A user downloads a fake application that contains malicious software and may steal data from their device.	Install trusted applications, update software, and use reputable security protection.
5	Password Attack	An attacker tries stolen or reused passwords to gain unauthorized access to an Instagram or Paytm account.	Use unique, strong passwords and enable two-factor authentication (2FA).

Conclusion

Network threats can lead to data theft, account compromise, and service disruption. Strong passwords, security updates, safe browsing, and network monitoring help reduce these risks.

2. Simulate a Brute-Force Attack Using Python

Python Code:

```
# Educational brute-force simulation

preset_pin = "2584"
attempts = 0

for number in range(10000):
guessed_pin = f"{number:04d}"
attempts += 1

if guessed_pin == preset_pin:
print("PIN guessed successfully:",
guessed_pin)
print("Number of attempts:", attempts)
break
```

Sample Output:

```
PIN guessed successfully: 2584
Number of attempts: 2585
```

Explanation:

This program simulates a brute-force attack by checking every possible four-digit PIN, starting from `0000`. The `for` loop generates each combination, and the `attempts` variable counts how many guesses have been checked. When the generated PIN matches the preset PIN, the

program displays the correct PIN and total number of attempts, then stops.

Security Lesson:

Brute-force attacks can guess weak PINs by trying many combinations. Strong authentication, attempt limits, temporary account lockouts, and multi-factor authentication help reduce this risk.

3. Physical Security Risks in a Server Room

A server room hosting an application like Zomato or BookMyShow contains important servers, network devices, and sensitive data. Physical security is necessary to protect this equipment from damage, theft, and unauthorized access.

No.	Physical Security Risk	Possible Impact	Mitigation
1	Unauthorized Access	An unauthorized person could access servers, steal equipment, or damage network devices.	Install biometric access control or smart card locks and allow entry only to authorized staff.
2	Fire	A fire could damage servers, switches, routers, and storage devices.	Install smoke detectors, suitable fire suppression systems, and fire alarms.
3	Power Failure	A sudden power outage could interrupt services and cause unexpected system shutdowns.	Install an Uninterruptible Power Supply (UPS) and a backup generator where required.
4	Water Leakage or Flooding	Water could damage electrical equipment and cause network downtime.	Install water-leak sensors, keep equipment away from water pipes, and maintain proper drainage.

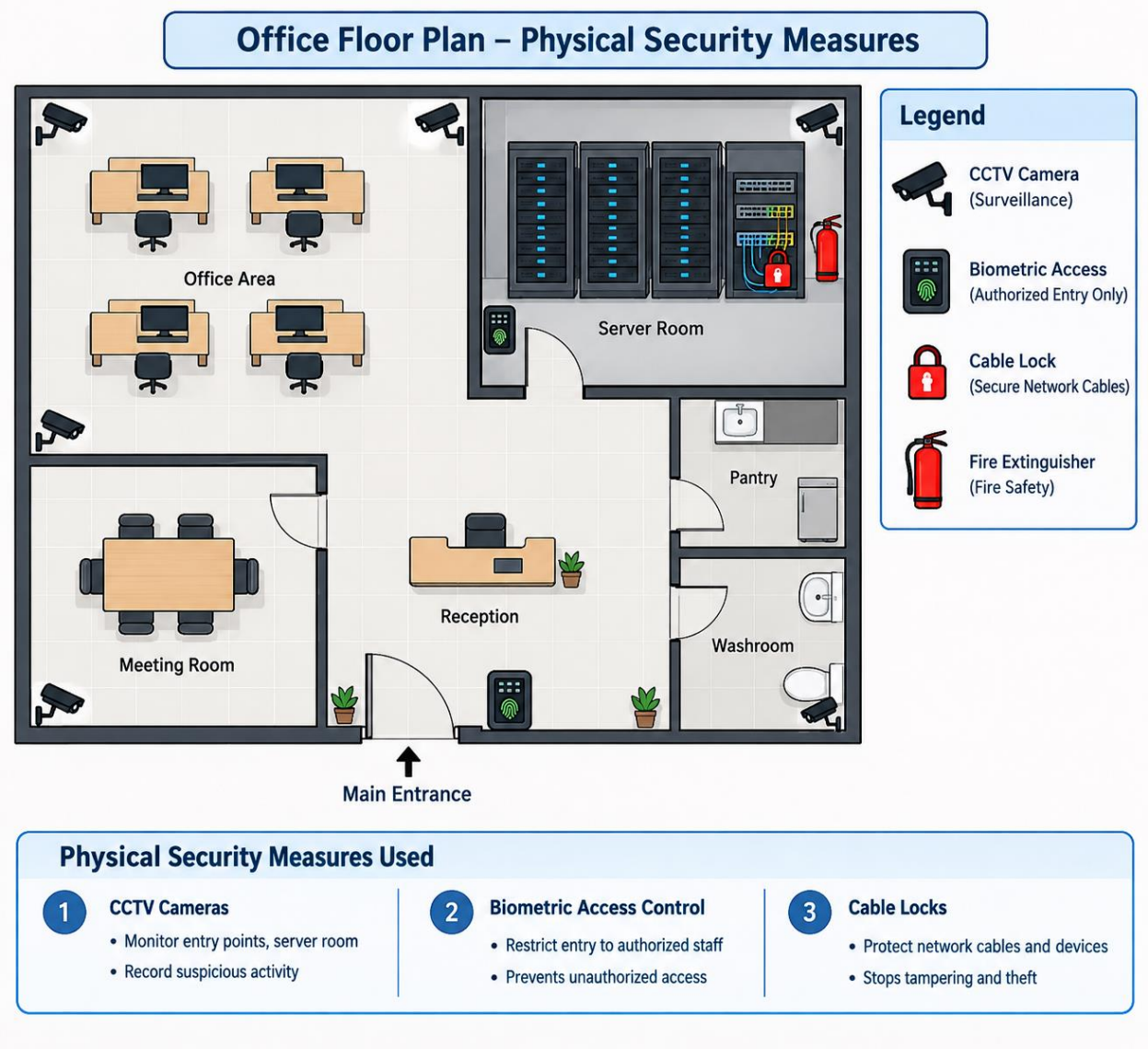
Conclusion

Physical security is essential for protecting server rooms and maintaining service availability. Access control, fire protection, backup power, and water-leak detection help reduce risks and protect critical IT infrastructure.

4. Draw a Floor Plan of a Small Office and Mark Physical Security Measures

Objective:

To design a small office floor plan and identify physical security measures used to protect network equipment from unauthorized access, theft, and damage.



Physical Security Measures Used:

1. **CCTV Cameras:** Monitor the office entrance and server room to detect suspicious activities.
2. **Biometric Access Control:** Allow only authorized staff to enter the server room.
3. **Cable Locks:** Protect network equipment and cables from theft and tampering.
4. **Fire Extinguisher:** Help control fire-related emergencies and protect office equipment.

Conclusion:

These physical security measures help protect network equipment, prevent unauthorized access, and reduce the risk of theft and damage.

5. Checklist for Mitigating Network Threats and Physical Security Risks in a College Computer Lab

A network administrator must follow security practices to protect computers, network devices, user accounts, and important data. The following checklist is customized for a college computer lab.

Security Checklist

- **1. Update Systems and Antivirus:** Regularly update lab computers, operating systems, applications, and antivirus software to protect against malware and known vulnerabilities.
- **2. Use Strong Authentication:** Require strong passwords for administrator accounts, enable multi-factor authentication where available, and restrict access to authorized staff.
- **3. Configure Firewalls:** Enable host-based firewalls on lab computers and configure network firewalls to block unnecessary and unauthorized connections.
- **4. Secure Network Equipment:** Keep routers, switches, and servers in a locked cabinet or restricted server room to prevent unauthorized physical access.
- **5. Install CCTV and Access Control:** Monitor the computer lab entrance and network equipment area with CCTV cameras and use authorized entry procedures.
- **6. Protect Data and Create Backups:** Back up important student records, assignment files, and lab configurations regularly. Store backups securely and test restoration when needed.
- **7. Monitor and Report Security Incidents:** Review security logs, check for suspicious network activity, inspect cables and devices for tampering, and report incidents to the college IT administrator immediately.

Review and Customization

This checklist has been adapted specifically for a college computer lab. It focuses on shared computers, student accounts, lab network devices, academic files, and physical access. The network administrator should review these measures regularly and update them whenever new risks arise.

Conclusion

Following this checklist helps a college computer lab reduce network threats, protect equipment, maintain data confidentiality, and provide a safer computing environment for students and staff.